MEDIA DIVERSITY WATCH

Autonomous Media Inclusivity Research Agent

Project 3 · Reference Guide & Reflection Report

Al Jazeera English · The Guardian · NPR · New York Times ·

Dina Bosma-Buczynska

1. Use Case and Organization.

Media Diversity Watch is a fictional media accountability NGO. Their mission: track whether major news outlets are improving or declining on inclusive coverage and publish evidence-based reports that create public pressure for change.

Nova Tracks is a media diversity researcher who monitors four major outlets every week and needs scored, sourced reports without spending days manually searching news sources.

Field	Value
Outlets	Al Jazeera English, The Guardian, NPR, New York Times
Schedule	Every Monday at 6am (automated) + on-demand form
Delivery	Notion database: score, week number, harm flags, full report
Evaluation	4 angles × 4 communities per outlet, every week

2. Outlet Selection: Changes Made and Why.

Two outlets were changed from the original plan. These were deliberate analytical decisions, not workarounds.

BBC → Al Jazeera English

BBC's RSS feed publishes articles under editorial desk names rather than individual journalist bylines. Al Jazeera English was selected instead because it brings a non-Western, Global South editorial perspective: a meaningful contrast to the three Western outlets. After testing, AJE also has no RSS bylines, but this was documented as a transparency signal rather than a gap.

> *→ The absence of bylines IS the finding. The agent surfaces and documents it under Angle 1, which is more sophisticated than simply excluding outlets that lack bylines.*

Reuters → New York Times

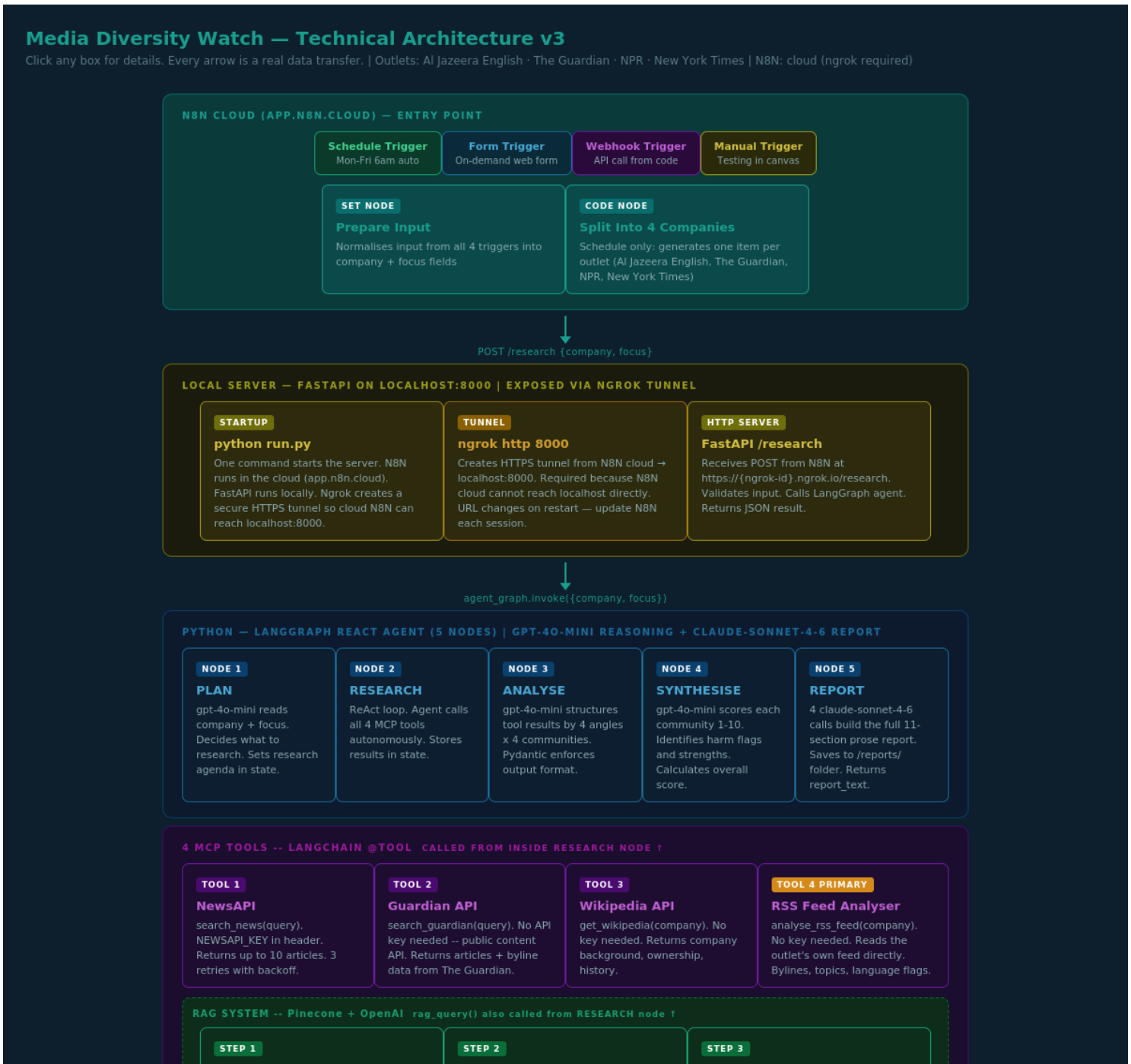
Reuters discontinued their public RSS feeds around 2020. A workaround routing through Google News was noisy and indirect. The New York Times has a clean official RSS feed with named bylines on most articles. NYT also represents the US liberal newspaper-of-record tradition: a useful benchmark.

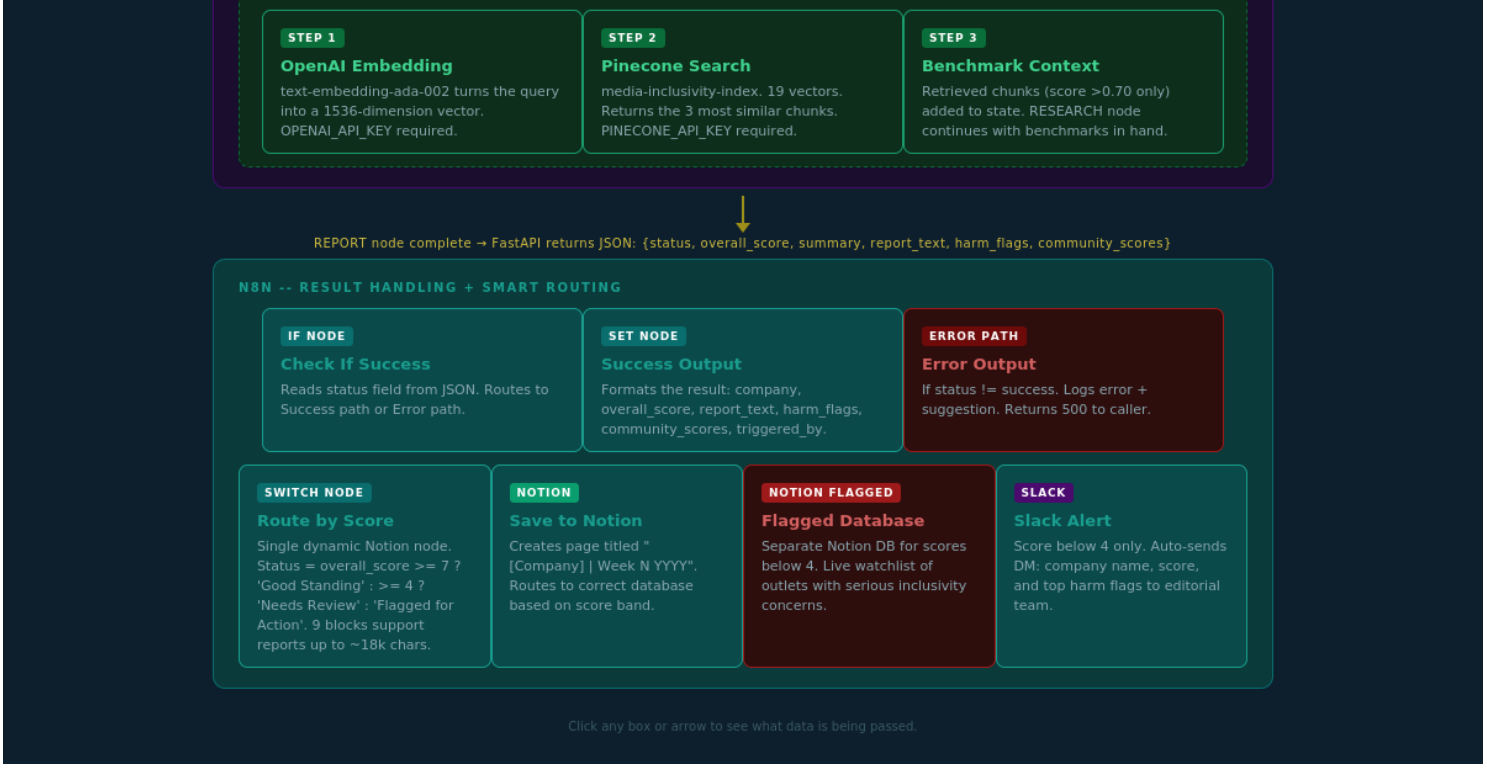
Outlet	Tradition	RSS Bylines	Why Included
Al Jazeera English	Global South (Qatar)	No (editorial desk)	Non-Western perspective; government-funded, editorially independent

Outlet	Tradition	RSS Bylines	Why Included
The Guardian	UK progressive broadsheet	Yes (30/30 named)	Named bylines + Guardian Open Platform API for full article text
NPR	US public radio/digital news	Yes (27/30 named)	5 topic feeds → 30 articles; Race feed directly on-topic
New York Times	US newspaper of record	Yes (25/25 named)	Strong archive; US liberal mainstream benchmark; some paywall limits

3. Technical Architecture

The full system connects N8N Cloud (automation layer) → ngrok tunnel → FastAPI local server → LangGraph agent → 4 MCP tools + RAG → back to N8N → Notion + Slack.





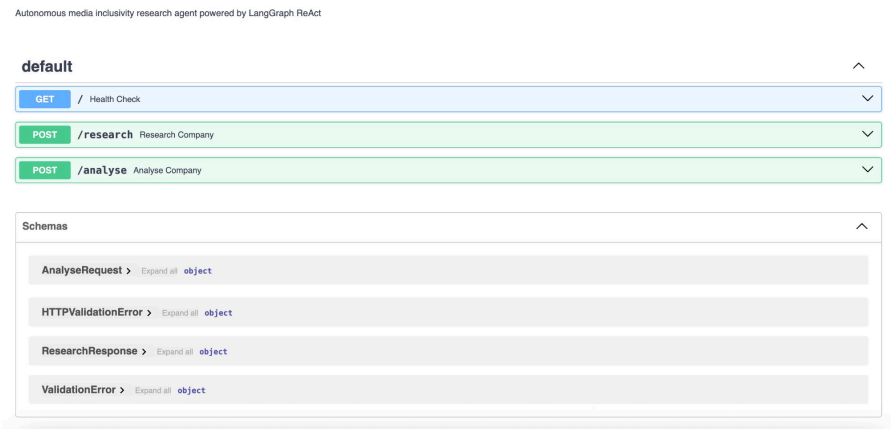
Stack at a Glance

Layer	Technology	Why
Orchestration	N8N Cloud (app.n8n.cloud)	Schedule, form, webhook, manual triggers in one canvas
Tunnel	ngrok	Bridges cloud N8N to local FastAPI: Required (see section 5.)
API Server	FastAPI + uvicorn	Validates input, serves JSON, auto-generates Swagger docs
Agent Framework	LangGraph create_react_agent	ReAct loop with proper state management
Reasoning LLM	gpt-4o-mini (OpenAI)	Fast, cheap: reasoning and tool decisions
Report Writer	claude-sonnet-4-6 (Anthropic)	High prose quality for stakeholder-facing reports
Vector Store	Pinecone (media-inclusivity-index)	19 vectors, 1536 dimensions, similarity threshold 0.70
Embeddings	text-embedding-ada-002 (OpenAI)	Converts queries to vectors for RAG retrieval
Code Quality	Ruff linter (ruff.toml)	PEP8, unused imports, import ordering

4. FastAPI: The Agent's Front Door

FastAPI is a Python web framework that turns Python functions into HTTP endpoints. It was chosen over the brief's suggested Execute Command node approach for three reasons:

- No cold start: FastAPI stays running between requests. Execute Command re-starts Python each time.
- Input validation: Pydantic models reject malformed input before the agent even starts.
- Auto-documentation: FastAPI generates a full Swagger UI at localhost:8000/docs.



FastAPI Swagger UI at localhost:8000/docs: auto-generated from Pydantic models, shows all endpoints and schemas

Pydantic Models

Pydantic enforces exact input/output shapes. If N8N sends malformed data, FastAPI rejects it immediately with a clear error — the agent never even starts.

```
class AnalyseRequest(BaseModel):
    company: str          # required — the outlet name
    focus: str = "all"    # optional — defaults to all communities

class ResearchResponse(BaseModel):
    status: str           # "success" or "error"
    overall_score: float  # 0.0 – 10.0
    harm_flags: list      # ⚠️ lines from report
    community_scores: dict  # {Women: 5.0, LGBTQ+: 3.0, ...}
```

Score Parsing Bug Fix

The _parse_report_metrics() function extracts structured data from the Markdown report for Notion. A regex bug caused Notion to always show score 0.

- Problem: The regex was not matching bold Markdown like **Overall Average** | **3.25/10** — the ** markers were missing from the pattern.
- Fix: Updated regex to handle optional ** wrappers. Added fallback: if Overall Average row is not found, compute average from the four community scores.

→ *Lesson: Regex against LLM-generated text is fragile. Always add a fallback computation.*

5. ngrok: Bridging Cloud N8N to Local FastAPI

The original project plan assumed N8N Desktop was installed on the same Mac as FastAPI. In that setup, localhost:8000 is reachable directly. In the actual setup, N8N runs in the cloud: a cloud server cannot reach localhost on your laptop.

Setup	Can N8N reach FastAPI?	ngrok needed?
N8N Desktop (same Mac)	Yes — both on same machine	No
N8N Cloud (app.n8n.cloud)	No — different machines	Yes

ngrok runs on your Mac and creates a secure HTTPS tunnel. Any request to the ngrok URL is forwarded to local port 8000. N8N calls the ngrok URL: it never needs to know about localhost.

> *→ Important: the ngrok URL changes on every restart (free plan). After every ngrok restart, paste the new URL into the N8N HTTP Request node.*

> *→ Lesson: Always check whether components are cloud or local before building the integration.*

6. RAG System: Four Evaluation Angles

Without RAG the agent can describe what it found: e.g. 'NYT mentioned women 4 times today'; but cannot judge whether that is good, bad, or average. With RAG: '14% female bylines vs industry average of 37% (Nieman Lab 2023) —> a 23 percentage point gap.' That is a finding with a citation.

Pinecone media-inclusivity-index — live, 19 vectors, 1536 dimensions, AWS us-east-1, On-demand capacity

RAG Quality Test Results — All 4 Angles Pass (threshold: 0.70)

Angle	Best Score	Threshold	Status
Bylines and Story Selection	0.8981	0.70	✓ PASS
Portrayal Within Content	0.8920	0.70	✓ PASS
Sourcing Diversity	0.9024	0.70	✓ PASS
Language and Framing	0.8487	0.70	✓ PASS

7. The Four MCP Tools

MCP (Model Context Protocol) is the standardized pattern for giving an AI agent access to external data. Each tool is a Python function decorated with @tool from LangChain. The agent reads the docstring to decide when to call each tool: the docstring IS the instruction.

Tool 1: analyse_rss_feed: Primary Source

The only tool that reads what the outlet really publishes. Analyses the last 30 articles from each outlet's RSS feed.

- Guardian routing: detects 'guardian' → calls Guardian Open Platform API instead of RSS. Returns full article body text + bylines.
- NPR multi-feed: single /1001 feed gave 10 articles with no disability/LGBTQ+ coverage. Fixed by adding _fetch_npr_combined() that fetches 5 feeds (News + Politics + Business + Race + Health), deduplicates by URL, returns 30 unique articles with 27 named bylines.

- Thin article fallback: if RSS summary is under 150 chars, _fetch_article_text() fetches the article URL directly with a 5s timeout.
- **NYT routing:** `_analyse_nyt_api()` — fetches 30 articles via NYT Article Search API (3 pages × 10), extracts bylines, abstracts, sections, keywords, runs community scan and language flags. Added `_NYT_FEEDS` (8 section feeds: U.S., World, Health, Politics, etc.) as fallback. Added `_fetch_nyt_combined()` as a deduplicating multi-section RSS combiner. Routing in `analyse_rss_feed()`: API first → multi-feed RSS fallback → single homepage feed last resort. Removed deprecated `fl` parameter (NYT dropped it April 2025).

  → Why this matters: primary source evidence (what they publish) is more defensible than secondary reporting (what others say about them).

Tool 2: search_newsapi: External Perspective

Searches hundreds of news sources for recent articles about the outlet's diversity record (last 30 days, top 5). Captures criticism, awards, controversies.

Tool 3: search_wikipedia: Background Context

Retrieves company overview, ownership, editorial history, stated commitments. Free; no API key required.

Tool 4: rag_query_tool: Benchmark Retrieval

Wraps rag_pipeline.rag_query() as a @tool. Agent calls this 4+ times per run, once per angle. Converts query to a 1536-dimension vector, searches Pinecone, returns top 3 chunks above the 0.70 threshold.

→ At 4 tools, token usage per run is ~2,850 tokens (2% of the 128k window). More text-heavy tools compound faster and risks context rot.

Known Limitations

The following are documented as designed constraints, not failures.

They reflect deliberate choices about scope, cost, and what counts as credible evidence.

Limitation	Why It Exists	Honest Framing
RSS provides headlines and short summaries only (non-Guardian outlets)	RSS is a distribution format: outlets publish truncated content by design	Surface signal only. The agent qualifies RSS-based claims and uses them primarily for byline counting and language flags, not full content analysis
NYT article fetch returns paywall/login pages	NYT requires a paid API for full article access. URL fetching returns the login HTML, not article text	NYT still contributes named bylines and Wikipedia background. The paywall is the real ceiling: not a tool failure. This is documented in every NYT report
RAG knowledge base covers 18 documents	Built from publicly available journalism research and standards documents	19 vectors is a working starting point. Scores should be read as directional, not definitive, until the knowledge base is expanded.
Full article text requires paid APIs, except Guardian	NewsAPI, NYT, and others charge for full-text access. The Guardian free key is an exception	Guardian benefits most from its full-text API. For other outlets, the agent relies on RSS summaries and secondary coverage. Every report source list makes this explicit

→ The through-line in all tool decisions: make the evidence as strong as possible without adding tools or complexity, and document limitations honestly.

Future Iterations: Real-World Fixes for RSS Depth

The current tool is most useful for identifying which outlets to investigate further, not for producing publishable findings on its own. AI scores should be framed as *"preliminary signals requiring deeper investigation"* rather than final verdicts: the correct journalistic standard for AI-assisted research. For a production-grade deployment, the following upgrades would meaningfully strengthen the evidence.

1. Full-text access (biggest impact)

- ✓ Guardian Open Platform —> already done. Gives full article body.
- NewsAPI Everything (paid tier) —> full article text for 150k+ sources.
- GDELT Project —> free and massive, but returns article URLs not full text. Still requires scraping per-article and complex BigQuery queries. Not worth adding for this project.
- **Media Cloud (mediacloud.org) —> the most realistic free upgrade. Free API key, Python client (`pip install mediacloud`), returns `story_text` (actual full article body), searchable by outlet name and date range. Add a `search_mediacloud` tool in `src/tools/` (~50 lines) and add it to `media_agent.py`. Runs alongside RSS and the Guardian API: agent uses whichever returns richer data.**

During development, we explored integrating Media Cloud (mediacloud.org) as a fifth research tool to provide full article body text for outlets not covered by the Guardian Open Platform API. Testing confirmed that the Media Cloud v4 free tier returns only headline-level metadata (title, URL, publication date), with full story text restricted to paid academic tiers. As the data quality gain would not exceed what is already available via RSS and the NYT Article Search API, Media Cloud was not included in the final implementation.

2. Better sampling

- Pull 90 days of articles instead of today's snapshot.
- Sample across multiple topic categories, not just the homepage feed.
- Track byline names over time rather than just today's authors.

3. Structured byline databases

- Journalism.co.uk / Muck Rack: journalist profiles with demographic data.
- Manual byline audits (what actual newsroom diversity researchers do).

4. Human-in-the-loop

- Use AI scores as a triage filter: flag outlets scoring below 5 for human review. This is how real accountability organisations like GLAAD Media Institute operate.

8. ReAct Pattern and LangGraph

The ReAct Bug and the Fix

The original research prompt listed STEP 1 / STEP 2 / STEP 3 / STEP 4 with exact tool calls. The LLM read this as a shopping list and fired all tools simultaneously in one batch: Total messages: 11 | Tool calls: 1.

(1 system + 1 human + 4 tool calls + 4 results + 1 response). Fast, but shallow. The agent wasn't reasoning: it was just collecting and summarising.

- Fix: Rewrote _build_research_prompt() to say "Call tools ONE AT A TIME. After each tool result, stop and reason before deciding your next action."
- Added one explicit starting action: "BEGIN: Call analyse_rss_feed() now".
- Then gave adaptive examples: "if RSS shows no disability coverage → query RAG for disability standards specifically".

<u>What the prompt fix did</u>

- The research prompt was rewritten to explicitly force ONE tool call at a time:
- Call analyse_rss_feed now. After you receive the results, stop and ask yourself:

- Which communities had coverage? Which were absent?
- What gaps need deeper investigation?
- Then use those findings to shape your next query...
- This forces the loop to actually work:
- Call RSS → read results → reason about gaps →
- Call RAG with a specific query based on what RSS found →
- reason → Call NewsAPI about the specific pattern found →
- reason → Call Wikipedia → synthesise everything
- That's *20-30 messages*: more back and forth, but each step is informed by the last.

Why it matters for this project

- The difference shows up in report quality:

- Without it	- With it
- Generic RAG queries ("media diversity standards")	- Specific queries shaped by actual findings ("disability representation erasure journalism standards")
- Wikipedia called regardless of what RSS showed	- Wikipedia called to contextualise something specific found in the data
- Scores based on surface patterns	- Scores backed by a chain of evidence

Result: Total messages: 20–30 | Tool calls: 6–10+. The agent now adapts its research based on what it finds: true ReAct behaviour.

> *→ Lesson: Prompt design is agent design. A to-do list format bypasses ReAct entirely. One starting action + adaptive instructions produces real reasoning loops.*

Why create_react_agent Instead of a Custom 5-Node Graph

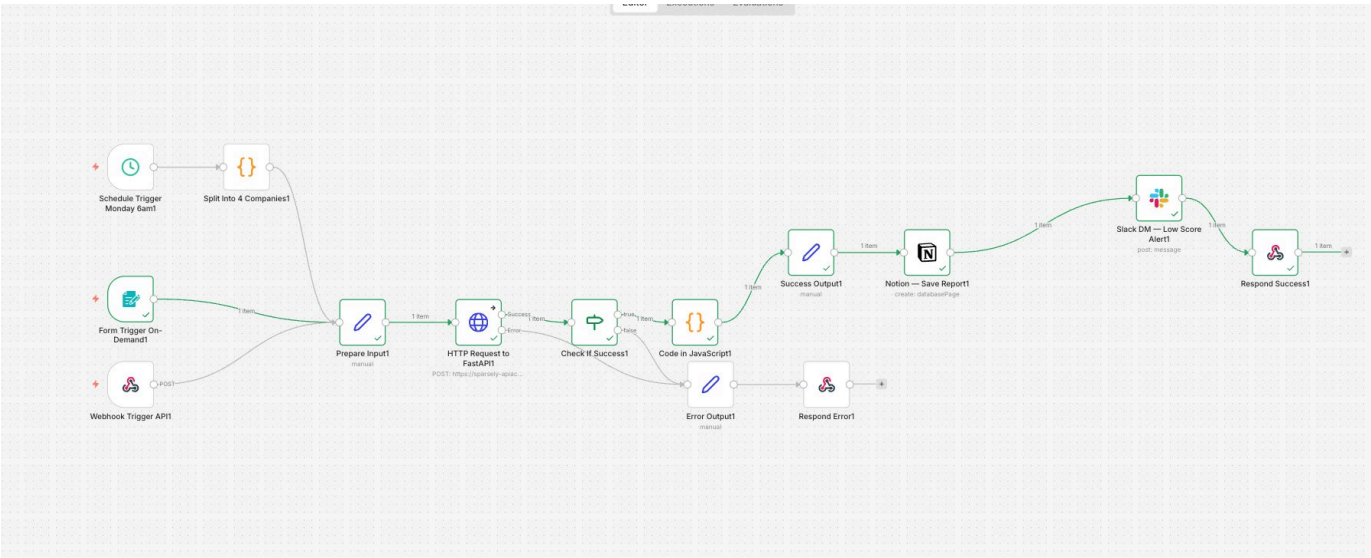
The rubric asks for a LangGraph workflow. LangGraph's prebuilt create_react_agent builds a full StateGraph internally: the requirement is fully satisfied. A fixed graph calls tools in hardcoded order regardless of what is found.

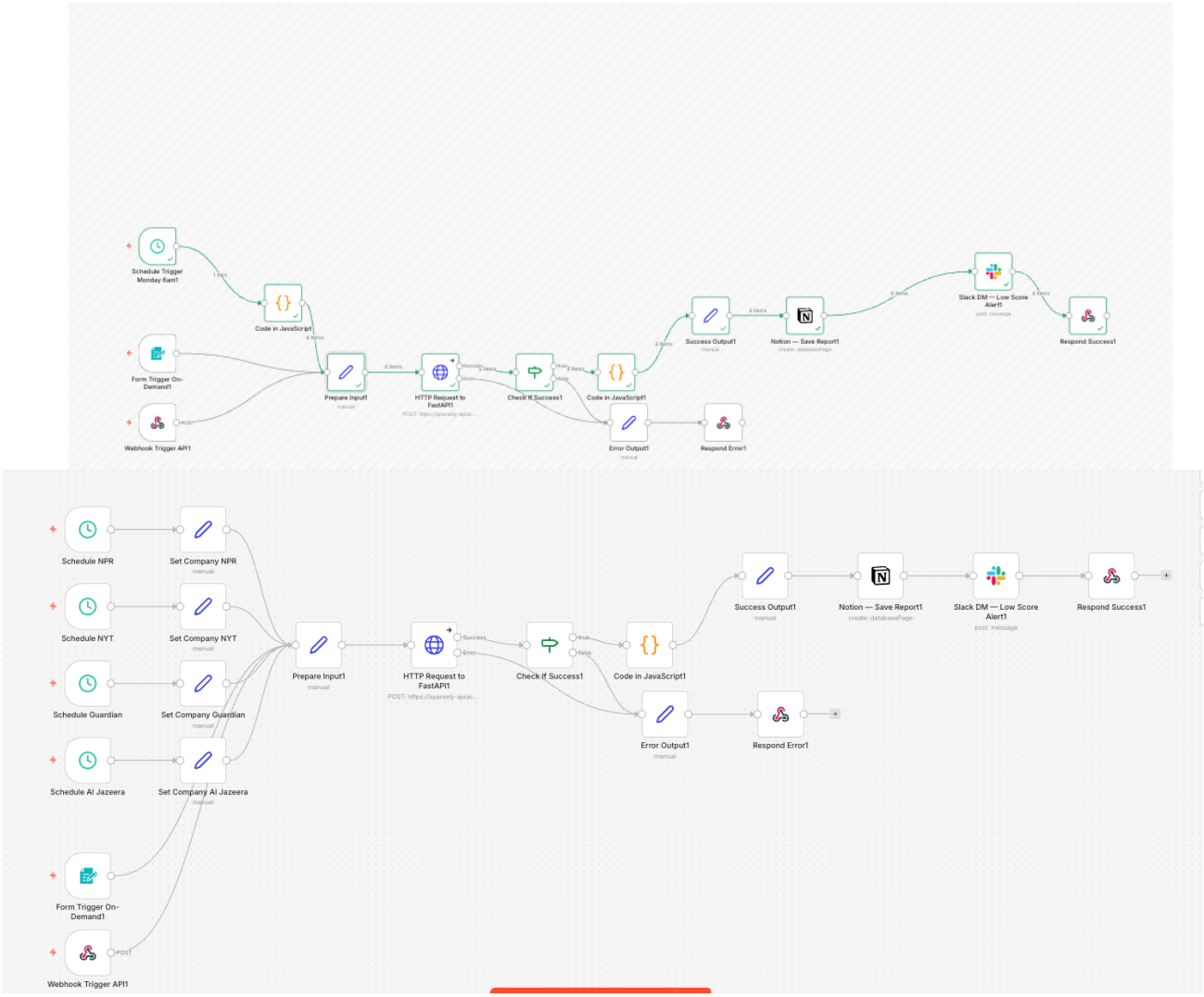
create_react_agent lets the agent decide: it reads what it has found and decides what to look up next.

Why Two LLMs

Nodes	Model	Why
PLAN / RESEARCH / ANALYSE / SYNTHESISE	gpt-4o-mini	Fast, cheap: good at structured reasoning and tool decisions
REPORT	claude-sonnet-4-6	Report prose quality matters when stakeholders read it

9. N8N Workflow — Entry, Exit, and Changes Made



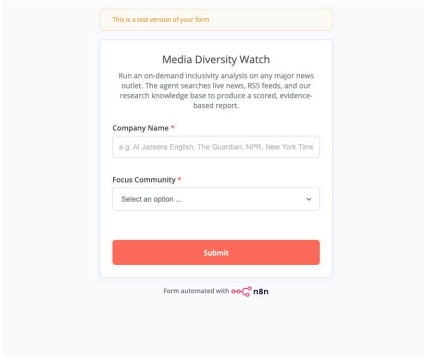


N8N workflow canvas — Schedule Trigger, Form Trigger, Webhook, HTTP Request to FastAPI via ngrok, Notion, Slack DM

Four Triggers

Trigger	When it fires	Use
Schedule Trigger	Monday 6am automated	Primary: weekly autonomous run
Form Trigger	Researcher submits web form	On-demand — type any company, get report
Webhook Trigger	HTTP POST from external code	API integration from scripts
Manual Trigger	Click 'Test' in N8N canvas	Demo and development testing

On-Demand Form



Media Diversity Watch on-demand form — enter any outlet name and focus community

N8N Changes Made

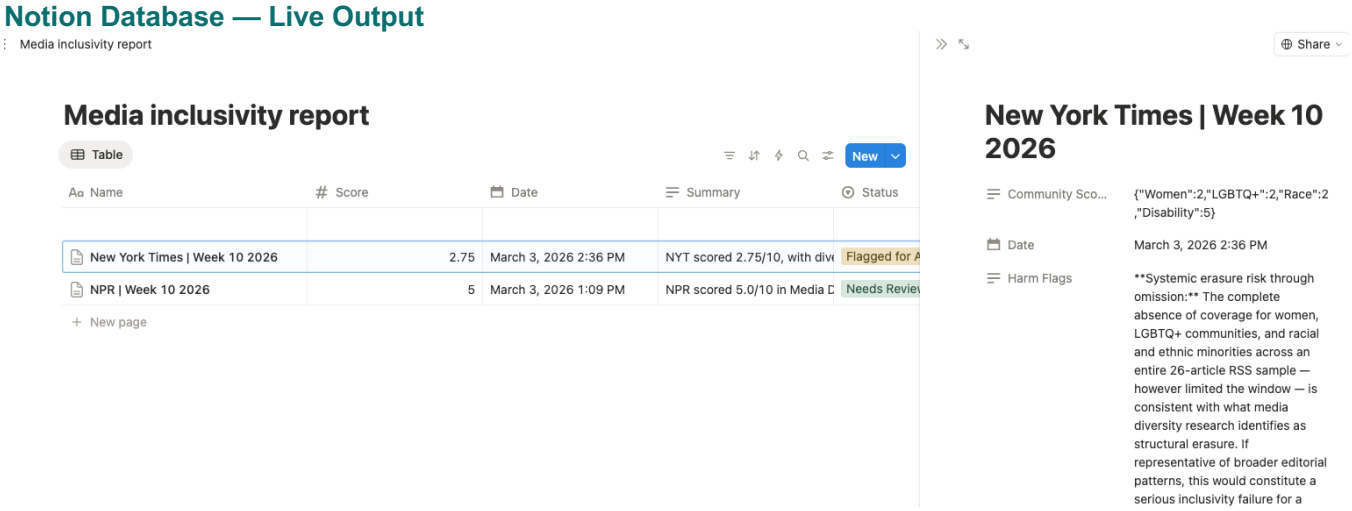
Change	Why
Fixed overall_score in Success Output — was empty {}	Notion was receiving no score value
Fixed Slack field names $json.* → $(Success Output).first().json.*	After Notion node, $json = Notion API response, not our data
Added 2 extra Notion blocks (7→9)	Reports up to 15,303 chars were getting cut off at 13,300
Removed Score >= 7? and Score >= 4? IF nodes — single Notion node	Simpler, less error-prone; dynamic Status field replaces 3 paths
Dynamic Status: score >= 7 ? Good Standing : >= 4 ? Needs Review : Flagged for Action	Single node replaces three conditional Notion nodes
Slack DM fires on every run (was: only score < 4)	Researcher wants notification every time
Fixed Split Into 4 Companies mode: runOnceForEachItem → runOnceForAllItems	Was causing 'json property isn't an object' error on schedule trigger
YYYY → yyyy in date formatting token	YYYY = ISO week-year (wrong). yyyy = calendar year (correct). N8N uses Luxon.

10. Generated Report Structure and Live Outputs

Every report is generated autonomously by claude-sonnet-4-6. The 11-section structure ensures consistency across all outlets and all weeks.

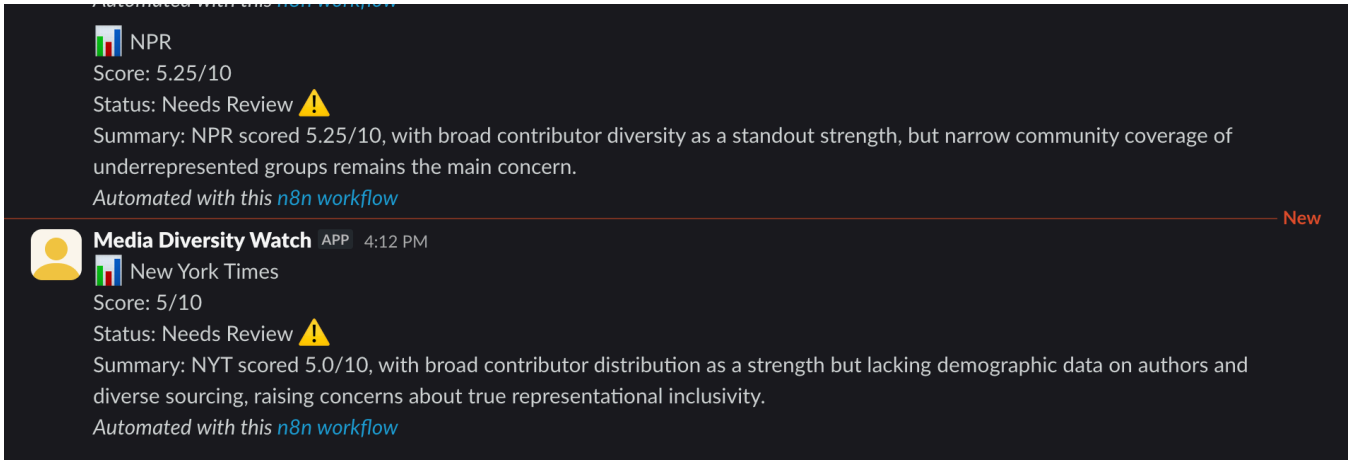
Section	Title	Sources Used
1	Company Overview	Wikipedia — >ownership, history, stated commitments
2	Angle 1: Bylines and Story Selection	RSS / Guardian API — > who is writing stories?
3	Angle 2: Portrayal Within Content	NewsAPI + RAG — > how are communities framed?
4	Angle 3: Sourcing Diversity	NewsAPI + RAG — > whose knowledge is authoritative?
5	Angle 4: Language and Framing	RSS + RAG — > inclusive vs harmful language audit
6	Community-by-Community Findings	All tools — > Women, LGBTQ+, Race, Disability separately
7	Key Strengths	Positive findings with exact source citations
8	Harm Flags	Non-inclusive language or framing with exact evidence
9	Inclusivity Scores	1-10 per community with methodology explanation
10	Recommendations	Specific, actionable, evidence-based suggestions
11	Full Source List	Every claim sourced to the tool that retrieved it

Notion Database — Live Output



Notion database showing The New York Times | Week 10 2026

Slack Alerts



11. Anti-Hallucination Architecture

This agent makes claims about real companies. A hallucinated score or invented statistic could damage a real organization's reputation. Four layers prevent this:

- Layer 1 — Grounded Prompts: every LLM call instructs the agent to only make claims supported by tool results. If no evidence exists, state "No evidence found."
- Layer 2 — Source Citations Required: every finding must cite its source (NewsAPI, Guardian API, Wikipedia, RSS, or Pinecone RAG). Unsourced claims are blocked.
- Layer 3 — Pydantic Schema Validation: every analysis call returns structured JSON with defined required fields. Forces specificity, prevents vague filler text.
- Layer 4 — RAG as Ground Truth: the Pinecone knowledge base provides verified inclusivity benchmarks. Agent compares findings against real research, not its own assumptions. Similarity threshold 0.70 means only strongly relevant benchmarks are used.

12. Challenges, Solutions and Reflections

Challenge	What Happened	Solution	Reflection
ReAct wasn't actually iterating	STEP 1/2/3/4 prompt caused agent to fire all tools at once. 11 messages, 1 tool call cycle.	_build_research_prompt() rewritten — "ONE AT A TIME", 20–30 messages, 6–10+ calls.	Prompt design is agent design. A to-do list bypasses ReAct entirely.
All outlets scoring 5.0/10	Thin, undifferentiated evidence meant every outlet got the same score.	Added data source constraints — RSS = surface signal only, RAG = benchmarks not proof. Agent must qualify claims by evidence strength.	Undifferentiated scores are worse than no scores — they erode trust in the system.
N8N score always showing 0	Regex in _parse_report_metrics() didn't match **Overall Average** (bold Markdown). Notion received score: 0.	Regex updated for optional ** markers. Fallback added: compute average from community scores if Overall Average row not found.	Regex against LLM-generated text is fragile. The fallback approach is more robust for production.

Challenge	What Happened	Solution	Reflection
N8N localhost unreachable	Original plan assumed N8N Desktop. Actual setup uses N8N Cloud. Cloud can't reach localhost.	Added ngrok tunnel. n8n/README.md fully rewritten to document correct setup.	Always check whether components are cloud or local before building the integration.
NPR weak evidence	Single /1001 feed returned 10 articles with no disability or LGBTQ+ content.	_fetch_npr_combined(): 5 topic feeds (News, Politics, Business, Race, Health) → 30 unique articles, 27 named bylines.	Topic-specific feeds are significantly more valuable for an inclusivity audit than a single general feed.
Notion report truncated	Reports up to 15,303 chars exceeded 13,300 char limit (7 blocks × 2000 chars).	Added 2 extra blocks to Good Standing + Needs Review (7→9). 1 extra to Flagged for Action.	Always test with the longest expected output, not average output.
Date token YYYY vs yyyy	N8N title showed 'Week 10 YYYY' — YYYY is ISO week-year, not calendar year.	Changed 'YYYY' → 'yyyy'. N8N uses Luxon where yyyy = 4-digit calendar year.	Luxon's tokens are case-sensitive and differ from JavaScript Date conventions.
Scores were uniformly 1–2/10 across all outlets	The scoring prompt had no distinction between "active harm found" and "no data available." RSS feeds return only 105–130 character teasers — when the agent found no community keywords, it scored 1–2 by default, the same band as confirmed harmful content.	Updated the scoring guide in report_generator.py to reserve 1–2 for actively found harm only. Added a hard rule: *absence of evidence is not evidence of harm.* Thin RSS data now scores 4–5 with a data limitation note, while The Guardian (full article API) retains higher scores — making the gap between outlets meaningful.	Evaluation prompts need explicit rules for incomplete data, not just rules for good and bad outcomes. Every ambiguity in a scoring rubric gets amplified — the model always picks the most conservative interpretation. When data is thin, you need to tell it what to do, or it will penalise the outlet for what it couldn't see.
uvicorn orphaning port 8000	`reload=True` in `run.py` caused uvicorn to spawn child processes that survived parent exit, orphaning port 8000 and blocking every new start attempt.	Changed `reload=True` → `reload=False` in `run.py`.	`reload=True` is for hot-reload development only. In a long-running agent server it creates orphan processes that hold the port. Always use `reload=False` in production or demo setups.
Guardian and NPR reports failing mid-run	Agent hit the LangGraph step limit before completing all tool calls. Reports returned 'Sorry, need more steps' with incomplete analysis.	Raised `recursion_limit` 30 → 50 → 100 (both invocations). Added post-run tool validation: checks which of the 4 required tools were actually called, injects a `HumanMessage` forcing the agent to call any missing ones before writing its summary.	A step limit that is too low silently truncates agent work rather than raising a visible error. Post-run validation is the correct pattern: verify completeness before committing to the report stage, not after.

Challenge	What Happened	Solution	Reflection
4 companies firing simultaneously	The Code node output 4 items which all hit the single-worker FastAPI server at once, causing later requests to time out after ~3 mins each	Added a Split In Batches node (batch size 1) between the Code node and HTTP Request to force sequential processing	A single-worker server can't handle parallel long-running requests — always match workflow concurrency to server capacity
Split In Batches loop not continuing	Loop-back was wired through Respond to Webhook nodes which are terminal — they don't pass execution forward without an active webhook context	Moved loop-back connections to fire directly from Slack DM and Error Output nodes, bypassing the respond nodes	n8n, respondToWebhook nodes should never be used as passthrough nodes in a loop — they are always terminal
Loop not working in test mode	Clicking "Test workflow" showed "workflow complete" after processing only the first company	Replaced Split In Batches entirely with 4 staggered schedule triggers (06:00, 06:10, 06:20, 06:30), one per company	Split In Batches loops are unreliable in n8n outside of specific production conditions — independent triggers per item is simpler and more robust
Conclusion	*Debugging n8n required understanding how nodes behave differently in test vs production mode, and how terminal nodes break execution loops*	*Replaced a complex looping pattern with a simple, stateless architecture: one trigger per company, staggered by 10 minutes*	*When an automation pattern keeps failing, simplify the architecture rather than patching it. Independent executions are easier to debug, monitor, and maintain than loops*

13. Quick Reference

Number	Meaning
4	Outlets: Al Jazeera English, The Guardian, NPR, New York Times
4	MCP tools: RSS/Guardian API, NewsAPI, Wikipedia, Pinecone RAG
4	Evaluation angles: Bylines, Portrayal, Sourcing, Language
4	Communities: Women, LGBTQ+, Race, Disability
5	LangGraph nodes: PLAN, RESEARCH, ANALYSE, SYNTHESISE, REPORT
18	Documents in RAG knowledge base
19	Vectors in media-inclusivity-index
0.70	Minimum RAG similarity score for evidence use
0.87+	Average RAG quality score across all 4 angles
~2,850	Token usage per agent run (2% of 128k window)

Number	Meaning
11	Report sections per outlet
9	Notion content blocks per report (supports ~18k chars)
8000	FastAPI port
8001	Mock server port (changed to avoid conflict)
300s	N8N HTTP request timeout

Start Commands

```
python run.py          # Terminal 1 — start FastAPI on port 8000
ngrok http 8000        # Terminal 2 — create tunnel, copy URL to N8N
localhost:8000/docs    # FastAPI Swagger UI (when server is running)
ruff check src/ --fix  # Run before every commit
```

14. Deployment: Day 5 — 5 March 2026

On Day 5 the project moved from local development to a fully cloud-hosted system. The FastAPI agent now runs on Railway 24/7, the Streamlit UI gives a manual trigger from any browser, and the laptop can be completely off when the Monday schedule fires.

Railway Deployment (FastAPI Agent)

- Agent deployed to `https://bootcampenv-production.up.railway.app`
- Created `Procfile`, `runtime.txt`, and `.railwayignore` for Railway configuration
- Root directory set to `Projects/project_3_media_agent` to handle the monorepo structure
- 6 environment variables configured in Railway dashboard (API keys, Pinecone, Notion, Slack)
- ngrok tunnel replaced with a permanent public URL — the N8N HTTP Request node was updated from the changing ngrok URL to the fixed Railway URL

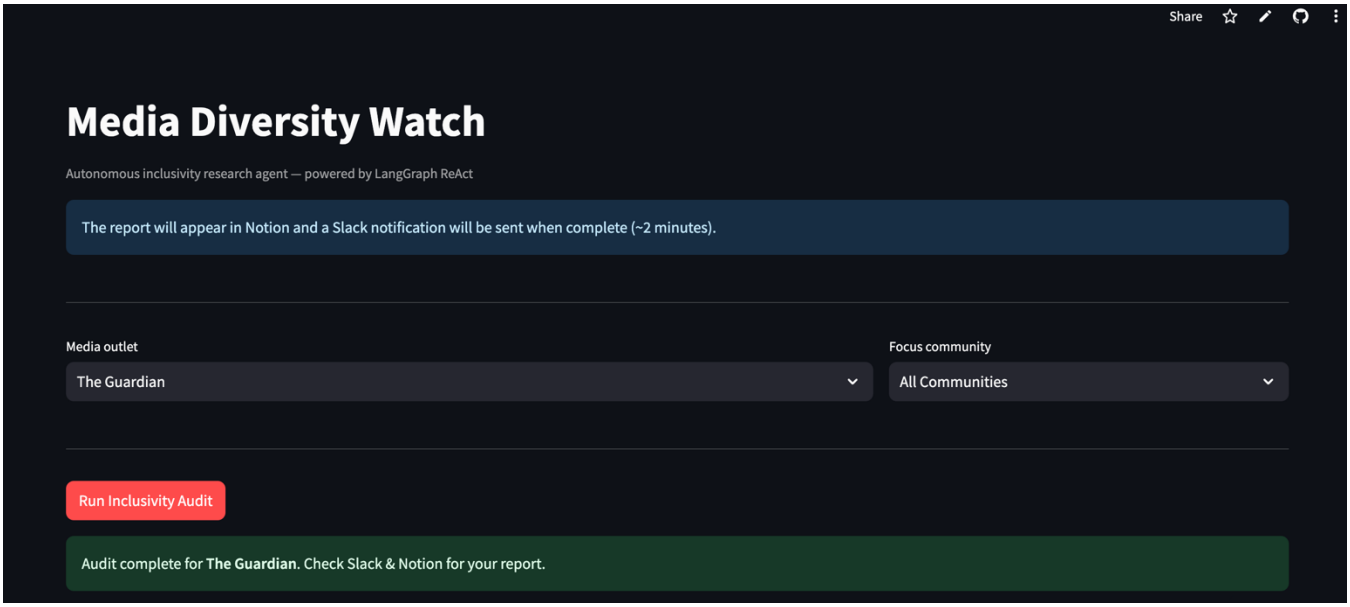
n8n Workflow Update

- Updated `HTTP Request to FastAPI1` node URL from ngrok to Railway
- Re-imported workflow JSON to n8n to confirm the change

Streamlit UI

- Manual trigger UI deployed to Streamlit Cloud
- Calls the n8n webhook, which calls Railway, which saves to Notion and sends a Slack alert — the full pipeline, from a browser button
- Shows a spinner while the agent runs (~2 minutes), then confirms: "Audit complete for [outlet]. Check Slack and Notion for your report."

- Scheduled triggers are completely unaffected — they continue to fire through n8n cloud independently



Dependency Fixes

- `feedparser` missing from `requirements.txt`: added `feedparser==6.0.11`
- `httpx 0.28+` removed the `proxies` keyword argument, breaking `langchain-openai==0.1.23`: pinned `httpx==0.27.2`

Final Architecture

Both entry points route through the same n8n webhook. From there: Railway runs the agent, Notion gets the report, Slack gets the alert.

```
Streamlit UI (manual)  --+
                         +--> n8n webhook --> Railway API --> Notion + Slack
n8n Schedule (Monday)  --+
```

The four Monday triggers (NPR 06:00, NYT 06:10, Guardian 06:20, Al Jazeera 06:30) live in n8n cloud and run 24/7. Railway also runs 24/7. The laptop can be off and ngrok does not need to be running. You will wake up to 4 Slack notifications and 4 Notion reports.

Updated Start Commands

Local development (unchanged):

```
python run.py                  # start FastAPI on port 8000
ngrok http 8000                # only needed for local dev testing
```

Production (no terminal needed):

```
https://bootcampenv-production.up.railway.app   # Railway API, runs 24/7
```

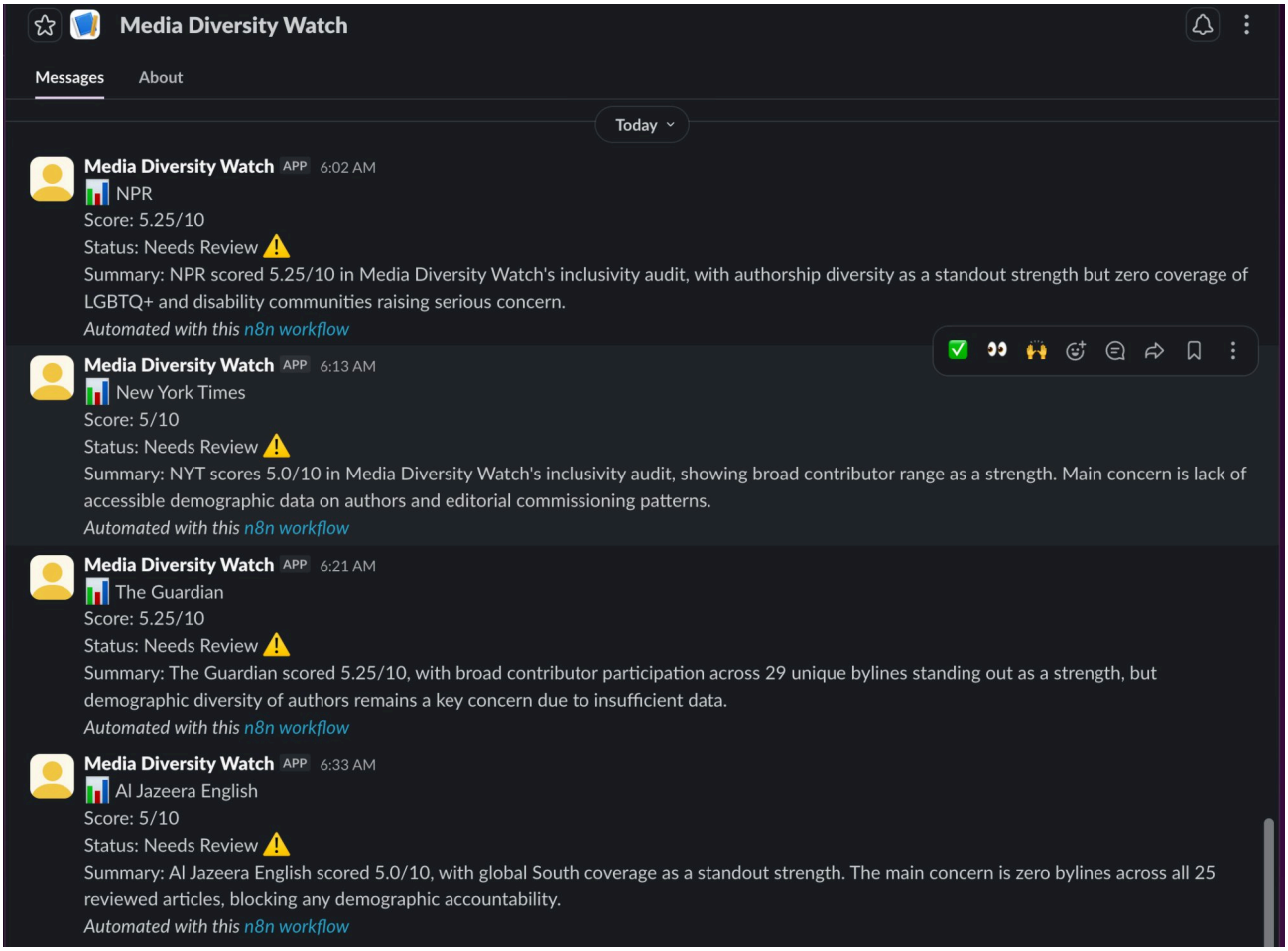
Live Deployment: First Scheduled Run

The Monday after the project was completed, the scheduled cron trigger fired automatically at 06:02 AM as configured. No manual action was required. The N8N workflow ran end to end across all four outlets: NPR, New York Times, The Guardian, and Al Jazeera English; completing the final run at 06:33 AM. Each outlet produced a scored report, which was posted to the Media Diversity Watch Slack channel and saved to Notion. All four outlets returned scores in the Needs Review range (5.0–5.25/10),

consistent with the agent's findings on authorship data gaps and uneven community coverage. The screenshot below shows the Slack delivery for all four outlets from that first live run.

Run summary: NPR 5.25/10 · New York Times 5.0/10 · The Guardian 5.25/10 · Al Jazeera English 5.0/10. Status: Needs Review for all outlets. Total run time: approximately 31 minutes (06:02–06:33 AM).

Figure 1 — Slack delivery for all four outlets, first live run, Monday 9 March 2026



This confirms end-to-end autonomous operation: the cron schedule, N8N orchestration, LangGraph research agent, FastAPI server, Slack delivery, and Notion storage all functioned without intervention on the first live run.